

Department of Computer Science and Engineering

Subject Name: **ARTIFICIAL INTELLIGENCE**

Subject Code: **CS T71**

Introduction: History of AI - Intelligent agents – Structure of agents and its functions - Problem spaces and search - Heuristic Search techniques – Best-first search - Problem reduction - Constraint satisfaction - Means Ends Analysis.

2 Marks

UNIT 1

TWO MARKS

1. Define the term Artificial Intelligence.

The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense.

The automation of activities that we associate with human thinking, activities such as decision-making, problem solving, learning.

2. Define an ‘Intelligent agent’?

An agent is anything that can be viewed as **perceiving** its environment through **sensors** **and acting** upon that environment through **actuators**.

3. what are the four types of agent program?

- Simple reflex agents
- model based agent
- Goal-based agents
- Utility-based agents

4. Define search?

In general, an agent with several immediate options of unknown value can decide what to do by first examining different possible sequences of actions that lead to states of known value, and then choosing the best sequence. This process of looking for such a sequence is called search.

5. Define breadth first search.

Breadth-first search is a simple strategy in which the root node is expanded first, then all the successors of the root node are expanded next, then their successors, and so on. In general, all the nodes are expanded at a given depth in the search tree before any nodes at the next level are expanded.

6. What is uniform cost search?

Breadth-first search is optimal when all step costs are equal, because it always expands the shallowest unexpanded node. By a simple extension, we can find an algorithm that is optimal with any step cost function. Instead of expanding the shallowest node, uniform-cost search expands the node n with the lowest path cost. Note that if all step costs are equal, this is identical to breadth-first search.

7. Define depth first search.

Depth-first search always expands the deepest node in the current fringe of the search tree. The search proceeds immediately to the deepest level of the search tree, where the nodes have no successors. As those nodes are expanded, they are dropped from the fringe, so then the search "backs up" to the next shallowest node that still has unexplored successors.

8. Define depth limited search.

The problem of unbounded trees can be alleviated by supplying depth-first search with a predetermined depth limit l . That is, nodes at depth l are treated as if they have no successors. This approach is called depth-limited search. The depth limit solves the infinite-path problem

9. What is bidirectional search?

The idea behind bidirectional search is to run two simultaneous searches—one forward from the initial state and the other backward from the goal, stopping when the two searches meet in the middle.

10. What is heuristic function?

A key component of informed searches is heuristic function, denoted by $h(n)$.

$h(n)$ = estimated cost of the cheapest path from node n to a goal node

11. What is A* search?

It evaluates nodes by combining $g(n)$, the cost to reach the node, and $h(n)$, the cost to get from the node to the goal.

$$f(n) = g(n) + h(n).$$

Since $g(n)$ gives the path cost from the start node to node n , and $h(n)$ is the estimated cost of the cheapest path from n to the goal, we have

$f(n)$ = estimated cost of the cheapest solution through n

12. What are the four components of learning agent?

The four components of learning agent are

Learning element

Performance element

Critic

Problem generator

13. What is recursive best first search?

It is a simple recursive algorithm that attempts to mimic the operation of standard best-first search.

It keeps track of the f-value of the best alternative path available from any ancestor of the current node. If the current node exceeds this limit, the recursion unwinds back to the alternative path. As the recursion unwinds, RBFS replaces the f-value of each node along the path with the best f-value of its children. In this way, RBFS remembers the f-value of the best leaf in the forgotten subtree and can therefore decide whether it's worth reexpanding the subtree at some later time.

14. Define iterative deepening depth first search

Iterative deepening search (or iterative deepening depth-first search) is a general strategy, often used in combination with depth-first search, that finds the best depth limit. It does this by gradually increasing the limit-first 0, then 1, then 2, and so on-until a goal is found. This will occur when the depth limit reaches d, the depth of the shallowest goal node.

15. Define Tic Tac Toe?

"Tic Tac Toe" and "Noughts and Crosses" redirect here. For Example consider two players, X and O, who take turns marking the spaces in a 3×3 grid

11 MARKS

1. What is an Intelligent agents and its environments?
2. Explain Structure of agents and its functions
3. Discuss Problem spaces and search
4. What is uninformed search strategies? Explain
5. Discuss Heuristic Search techniques - Best-first search
6. What is Problem Reduction? Explain it with with AO* Algorithm.
7. What is Constraint satisfaction? Explain
8. What is Means Ends Analysis.Explain
9. Discuss the History of AI .
- 10.Explain A* Algorithm.

UNIT – II

Knowledge Representation: Approaches and issues in knowledge representation- Propositional Logic – Predicate logic-Forward and backward reasoning - Unification- Resolution- Weak slot-filler structure – Strong slot-filler structure- Knowledge- Based Agent

UNIT 2

TWO MARKS

1. What is knowledge based agent?

The central component of a knowledge-based agent is. Its **knowledge base** or KB. Informally, a knowledge base is a set of representations of facts about the world. Each individual representation. SENTENCE" is called a **sentence** the sentences are expressed in a language called a **knowledge representation language**.

2. Describe Wumpus world?

The Wumpus world is a grid of squares surrounded by walls, where each Square can contain agents and objects. The agent always starts in the lower left corner, a square that we will label [1, 1]. The agent's task is to find the gold, return to [1, 1] and climb out of the cave.

3. Define Logic.

Logic consists of the following:

1. A formal system for describing states of affairs, consisting of
 - (a) The **syntax** of the language, which describes how to make sentences, and
 - (b) The **semantics** of the language, which states the systematic constraints on how sentences. Relate to states of affairs.
2. **The proof theory**—a set of rules for deducing the entailments of a set of sentences.

4. Define Propositional Logic.

In **propositional logic**, symbols represent whole propositions (facts); for example, *D* might have the interpretation "the wumpus is dead." which may or may not be a true proposition. Proposition symbols can be combined using **Boolean connectives** to generate sentences with more complex meanings.

5. Define First-Order Logic.

First-order logic commits to the representation of worlds in terms of **objects** and **predicates** on objects, as well as using **connectives** and **quantifiers**, which allow sentences to be written about everything in the universe at once. First-order logic seems to be able to capture a good deal of what we know about the world, and has been studied for about a hundred years

6. What is Unification?

Unification: The job of the unification routine, UNIFY, is to take two atomic sentences p and q and return a substitution that would make p and q look the same. (If there is no such substitution, then UNIFY should return *fail*.) Formally,

$\text{UNIFY}(p, q) = \theta$ where $\text{SuBST}(\theta, p) = \text{SuBST}(\theta, q)$

is called the **unifier** of the two sentences

7. What is Semantic Net?

A Semantic net is a representation of nodes and links. It describes how the representation is done in Artificial Intelligence.

8. Define Inference?

The terms "reasoning" and "inference" are generally used to cover any process by which conclusions are reached. In this chapter, we are mainly concerned with sound reasoning, which we will call **logical inference or deduction**. Logical inference is a process that implements the entailment relation between sentences.

9. What Is Resolution?

In mathematical logic and automated theorem proving, resolution is a rule of inference leading to a refutation theorem-proving technique for sentences in propositional logic and first-order logic. In other words, iteratively applying the resolution rule in a suitable way allows for telling whether a propositional formula is satisfiable and for proving that a first-order formula is unsatisfiable.

10. What Is Meant By Natural Deduction?

Natural deduction is a kind of proof calculus in which logical reasoning is expressed by inference rules closely related to the "natural" way of reasoning.

11. List Out The Issues In Knowledge Representation?

High level facts may not be adequate for inference substantial work must be done to reduce the knowledge into primitive form.

Low level primitives may require a lot of storage.

12. What is Universal Elimination?

Universal Elimination: For any sentence a , variable v , and ground term g :

$$\forall v a$$

$$\text{SUBST}(\{v/g\}, a)$$

For example, from $\forall x, \text{Likes}(x, \text{Ice Cream})$, we can use the substitution $\{x/\text{Ben}\}$ and infer $\text{Likes}(\text{Ben}, \text{Ice Cream})$.

13. What is Existential Elimination?

Existential Elimination: For any sentence a , variable v , and constant symbol k that does not appear elsewhere in the knowledge base:

$$\exists v a$$

$$\text{SUBST}(\{v/k\}, \alpha)$$

For example, from $\forall x \text{ Kill}(x, \text{Victim})$, we can infer $\text{Kill}(\text{Murderer}, \text{Victim})$, as long as *Murderer* does not appear elsewhere in the knowledge base.

14. Define Conceptual Dependency.

Conceptual Dependency originally developed to represent knowledge acquired from natural language input.

15. What is Knowledge Representation?

Knowledge representation is to express knowledge in computer-tractable Form, such that it can be used to help agents perform well. A knowledge representation language is defined by two aspects:

- The syntax
- The semantics

11 MARKS

1. What Is A Knowledge .How It Is Represented In Ai.
2. .What Is Knowledge Based Agents.Explain
3. Discuss Propositional Logic?
4. What Is Predicate Logic Or First Order Logic – Fol.Explain
5. What Is Unification. Explain
6. What Is Resolution?.Explain
7. Explain Weak Slot - Filler Structure
8. Explain Strong Slot - Filler Structure
9. Give A Detailed Explanation About Knowledge Representation and its Approaches And Issues?
10. Explain Forward And Backward Chaining?

UNIT – III

Reasoning under uncertainty: Logics of non-monotonic reasoning-Implementation- Basic probability notation - Bayes rule – Certainty factors and rule based systems-Bayesian networks – Dempster - Shafer Theory - Fuzzy Logic.

2 Marks

UNIT 3

Two Marks

1. Define The Term “Probability”.

To make a good decision, an agent cannot simply assume what the world is like and act according to those assumptions. It must consider multiple possible contingencies and their likelihood.

2. What Are Advantages Of Bayesian Network?

1. Specify a prior over world models.
2. Integrate using Bayes law with respect to all observed information to compute a posterior over world models.
3. Predict according to the posterior.

Interpolation Bayesian learning methods interpolate all the way to pure engineering.

Language Bayesian and near-Bayesian methods have an associated language for specifying priors and posteriors.

Intuitions Bayesian learning involves specifying a prior and integration, two activities which seem to be universally useful.

3. Define Probability Reasoning ?

The aim of a probabilistic logic (also probability logic and probabilistic reasoning) is to combine the capacity of probability theory to handle uncertainty with the capacity of deductive logic to exploit structure of formal argument. The result is a richer and more expressive formalism with a broad range of possible application areas. Probabilistic logics attempt to find a natural extension of traditional logic truth tables: the results they define are derived through probabilistic expressions instead.

4. What Is Non- Monotonic Reasoning?

The definite clause logic is monotonic in the sense that anything that could be concluded before a clause is added can still be concluded after it is added; adding knowledge does not reduce the set of propositions that can be derived.

A logic is non-monotonic if some conclusions can be invalidated by adding more knowledge. The logic of definite clauses with negation as failure is non-monotonic. Non-monotonic reasoning is useful for representing defaults. A default is a rule that can be used unless it is overridden by an exception.

5. Write Down Bayes' Theorem.

In probability theory and statistics, Bayes' theorem (alternatively Bayes' law or Bayes' rule) describes the probability of an event, based on conditions that might be related to the event. For example, suppose one is interested in whether Addison has cancer.

6. What is MYCIN?

MYCIN was an early expert system that used artificial intelligence to identify bacteria causing severe infections, such as bacteremia and meningitis, and to recommend antibiotics, with the dosage adjusted for patient's body weight — the name derived from the antibiotics themselves, as many antibiotics have the suffix "-mycin". The Mycin system was also used for the diagnosis of blood clotting diseases.

MYCIN was developed over five or six years in the early 1970s at Stanford University.

7. Explain Belief Network?

A belief network is a directed model of conditional dependence among a set of random variables. The precise statement of conditional independence in a belief network takes into account the directionality.

To define a belief network, start with a set of random variables that represent all of the features of the model. Suppose these variables are $\{X_1, \dots, X_n\}$. Next, select a total ordering of the variables, $X_1, \dots, X_n$.

8. What is the task of any Probabilistic inference system?

The Task of any Probabilistic inference system is to compute the posterior probability distribution for a set of query variables, given exact values of some evidence variable.

9. What is the tool that is used to deal with degree of belief?

The tool that is used to deal with the degree of belief is the probability theory which assigns a numeric degree of belief between 0 and 1.

10. What Is Meant By ATMS and JTMS ?

Assumption Truth Maintenance System

JTMS - Justification-Based Truth Maintenance System

11. What is a Utility Function?

The utility function $U(S)$

- An agent's preferences between different states S in the world are captured by the Utility function $U(S)$.
- If $U(S_i) > U(S_j)$ then the agent prefers state S_i before state S_j
- If $U(S_i) = U(S_j)$ then the agent is indifferent between the two states S_i and S_j

12. What are the two ways to represent Canonical Distributions?

1. Deterministic nodes
2. Noisy-OR relation

13. What are the four types of inferences?

1. Diagnostic Inferences
2. Casual Inferences
3. Intereasual Inferences
4. Mixed Inferences

14. Explain Node Ordering.

The correct order in which to add nodes is to add the root causes (parents first and then the variables) and the addition process is continued, until we reach the leaves which have no direct casual inference on other variables.

15. What Is Meant By Non-monotonic Logic ?

One system that provides a basis for default reasoning is Nonmonotonic logic in which the operator of first order logic is augmented with a modal operator M , which is read as "inconsistent".

11 MARKS

1. What Is Uncertainty. Explain.
2. What Is Non-Monotonic Reasoning .Explain
3. What Is Probability And Basic Probability Notation.
4. Explain Baye's Rule.
5. Explain Bayesian Networks.
- 6..Explain Dempster - Shafer Theory In AI.
7. Explain Fuzzy Logic In AI.
- 8.What Are The Advantages And Disadvantages Of Bayesian Network?

UNIT – IV

Planning and Learning: Planning with state space search - conditional planning-continuous planning - Multi-Agent planning. Forms of learning - inductive learning - Reinforcement Learning - learning decision trees - Neural Net learning and Genetic learning

2 Marks

UNIT 4

TWO MARKS

1. What is Planning in AI?

Planning is done by a planning agent or a planner to solve the current world problem. It has certain algorithms to implement the solutions for the given problem and this is called ideal planner algorithm

2. What is STRIPS?

STandford Research Institute Problem Solver

- Tidily arranged actions descriptions
- Restricted language (function-free literals)
- Efficient algorithms

3. What Is Partial Plan?

Partial plan is an incomplete plan which may be done during the Initial phase.

There are 2 main operation allowed in planning

- Refinement operator
- Modification operator
- (Partial) Plan consists of
- Set of operator applications S_i
- Partial (temporal) order constraints $S_i \rightarrow S_j$
- Causal links $S_i \xrightarrow{c} S_j$

4. What Is Conditional Planning?

Conditional planning Plan to obtain information (observation actions) Subplan for each contingency

Example: [Check(Tire1); If(Intact(Tire1); [In_ate(Tire1)]; [CallHelp])]

Disadvantage: Expensive because it plans for many unlikely cases

Similar to POP If an open condition can be established by observation action

5. Describe The Problem Generator.

It is responsible for suggesting actions that will lead to new and informative experiences. The point is that if the performance element had its way, it would keep doing the actions that are best, given what it knows. But if the agent is willing to explore a little, and do some perhaps suboptimal actions in the short run, it might discover much better actions for the long run.

6. What Is Meant By Speedup Learning?

The learning element is also responsible for improving the *efficiency* of the performance element. For example, when asked to make a trip to a new destination, the taxi might take a while to consult its map and plan the best route. But the next time a similar trip is requested, the planning process should be much faster. This is called **speedup learning**

7. What Are The Types Of Learning

- Supervised learning
- Unsupervised learning
- Reinforcement learning

8. What Are Decision Trees?

A **decision tree** takes as input an object or situation described by a set of properties, and outputs a yes/no "decision." Decision trees therefore represent Boolean functions.

Each internal node in the tree corresponds to a test of the value of one of the properties, and the branches from the node are labelled with the possible values of the test. Each leaf node in the tree specifies the Boolean value to be returned if that leaf is reached.

9. Define Learning .

Machine learning is a subfield of computer science that evolved from the study of pattern recognition and computational learning theory in artificial intelligence. In 1959, Arthur Samuel defined machine learning as a "Field of study that gives computers the ability to learn without being explicitly programmed".

10. What is mean by Reinforcement?

Reinforcement learning is an area of machine learning inspired by behaviorist psychology, concerned with how software agentsought to take actions in an environment so as to maximize some notion of cumulative reward. The problem, due to its generality, is studied in many other disciplines, such as game theory, control theory, operations research, information theory, simulation-based optimization, multi-agent systems, swarm intelligence, statistics, and genetic algorithms. In the operations research and control literature, the field where reinforcement learning methods are studied is called approximate dynamic programming. The problem has been studied in the theory of optimal control, though most studies are concerned with the existence of optimal solutions and their characterization, and not with the learning or approximation aspects

11. What Is Over fighting ?

Whenever there is a large set of possible hypotheses, one has to be careful not to use the resulting freedom to find meaningless "regularity" in the data. This problem is called overfitting. It is a very general phenomenon, and occurs even when the target function is not at all random. It afflicts every kind of learning algorithm, not just decision trees.

12. What Is Decision Tree Pruning?

Pruning works by preventing recursive splitting on attributes that are not clearly relevant, even when the data at that node in the tree is not uniformly classified.

13. What Is Pruning?

The probability that the attribute is really irrelevant can be calculated with the help of standard statistical software. This is called pruning.

14. What Is Cross Validation ?

Cross-validation is another technique that eliminates the dangers of overfitting. The basic idea of cross-validation is to try to estimate how well the current hypothesis will predict unseen data. This is done by setting aside some fraction of the known data, and using it to test the prediction performance of a hypothesis induced from the rest of the known data.

15. What Is Continuous Valued Attribute?

Attributes such as *Height* and *Weight* have a large or infinite set of possible values. They are therefore not well-suited for decision-tree learning in raw form. An obvious way to deal with this problem is to **discretize** the attribute. For example, the *Price* attribute for restaurants was discretized into \$, \$\$, and \$\$\$ values. Normally, such discrete ranges would be defined by hand. A better approach is to preprocess the raw attribute values during the tree-growing process in order to find out which ranges give the most useful information for classification purposes.

11 MARKS

1. What Is Planning In AI ? Explain The Planning Done By An Agent?
2. What Is Learning And Its Representation In Ai Explain?
3. Explain Inductive Learning
4. Explain Reinforcement Learning
5. Discuss Neural Net Learning.
6. What Is Conditional Planning? Explain.
7. Describe Continuous Planning.
8. Explain In Detailed About Forms Of Learning?
9. Discuss Genetic Learning.
10. Explain Multi Agent Planning.

UNIT – 5

External I/O Interfaces: Serial versus Parallel - USB: system, data transfer, and controller – Hot plugging – Low speed connections: RS232C and Parallel port: SPP, EPP, ECP – Local Area Networking: Requirements – Wired – Wireless – Bluetooth – Network Interface Cards – Wired Topologies – Switches/Access Points – Wireless Ethernet hardware – Network Protocols: IP and TCP/IP, IPX, NetBEUI. – Cables and Connections. Troubleshooting network problems.

PC Diagnostics, Testing & Maintenance: POST – Boot process – Maintenance tools – Preventive Maintenance.

2 Marks

Part 1

TWO MARKS

1. Define Expert System ?

An expert system is a computer program that simulates the thought process of a human expert to solve complex decision problems in a specific domain.

An *expert system* is an interactive computer-based decision tool that uses both facts and heuristics to solve difficult decision problems based on knowledge acquired from an expert.

2. What Are Applications Of Expert Systems?

- Interpreting and identifying
- Predicting
- Diagnosing
- Designing
- Planning
- Monitoring
- Debugging and testing
- Instructing and training
- Controlling

3. Define Robotics ?

"A reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks"

The branch of technology that deals with the design, construction, operation, and application of robots.

Robotics is a branch of engineering that involves the conception, design, manufacture, and operation of **robots**. This field overlaps with electronics, computer science, artificial intelligence, mechatronics, nanotechnology and bioengineering.

4. Write Some Applications Of Game Playing In AI?

In video games, artificial intelligence is used to generate intelligent behaviors primarily in non-player characters (NPCs), often simulating human-like intelligence. The techniques used typically draw upon existing methods from the field of artificial intelligence (AI). However, the term game AI is often used to refer to a broad set of algorithms that also include techniques from control theory, robotics, computer graphics and computer science in general.

5. Define Swarm Intelligence ?

Swarm intelligence research originates from work into the simulation of the emergence of collective intelligent behaviors of real ants. Ants are able to find good solutions to the shortest path problems between the nest and a food source by laying down, on their way back from the food source, a trail of an attracting substance—a pheromone. Based on the pheromone level communication, the shortest path is considered that with the greatest density of pheromone and the ants will tend to follow the path with more pheromone.

6. What Is Ant Algorithm?

The ant colony optimization algorithm (ACO) is a probabilistic technique for solving computational problems which can be reduced to finding good paths through graphs.

7. What Is Knowledge Acquisition?

Knowledge acquisition is the process used to define the rules and ontologies required for a knowledge-based system. The phrase was first used in conjunction with expert systems to describe the initial tasks associated with developing an expert system, namely finding and interviewing domain experts and capturing their knowledge via rules, objects, and frame-based ontologies.

Expert systems were one of the first successful applications of artificial intelligence technology to real world business problems.

8. What Do you Mean By Alpha Beta Pruning?

Alpha–beta pruning is a search algorithm that seeks to decrease the number of nodes that are evaluated by the minimax algorithm in its search tree. It is an adversarial search algorithm used commonly for machine playing of two-player games (Tic-tac-toe, Chess, Go, etc.).

It stops completely evaluating a move when at least one possibility has been found that proves the move to be worse than a previously examined move. Such moves need not be evaluated further.

When applied to a standard minimax tree, it returns the same move as minimax would, but prunes away branches that cannot possibly influence the final decision

9. What Is Expert System Shell ?

A rule-based, expert system maintains a separation between its Knowledge-base and that part of the system that executes rules, often referred to as the **expert system shell**. The system shell is indifferent to the rules it executes

10. What Is An Effectors?

Effectors are the means by which robots move and change the shape of their bodies. To understand the design of effectors,, it will help to talk about motion and shape in the abstract, using the concept of a degree of freedom(DOF).

11. . What Is A Sensor?

Sensors are the perceptual interface between robots and their environments. Passive sensors. Such as cameras are true observers of the environment: they capture signals that are generated by other sources in the environment. Active sensors, such as sonar, send energy into the environment

12. Write short notes on Heuristic Reasoning?

A heuristic is a mental shortcut used to solve a particular problem; it is a quick, informal, and intuitive algorithm your brain uses to generate an approximate answer to a reasoning question.

A heuristic technique often called simply a heuristic, is any approach to problem solving, learning, or discovery that employs a practical method not guaranteed to be optimal or perfect, but sufficient for the immediate goals.

13. What is meta data in Knowledge base

An expert system is, typically, composed of two major components, the Knowledge-base and the Expert System Shell. The Knowledge-base is a collection of rules encoded as **metadata** in a file

14. List The Knowledge System Building Tools ?

1. **Knowledge base:** A store of factual and heuristic knowledge. An ES tool provides one or more knowledge representation schemes for expressing knowledge about the application domain. Some tools use both frames (objects) and IF-THEN rules. In PROLOG the knowledge is represented as logical statements.
2. **Reasoning engine:** Inference mechanisms for manipulating the symbolic information and knowledge in the knowledge base to form a line of reasoning in solving a problem. The inference mechanism can range from simple *modus ponens* backward chaining of IF-THEN rules to case-based reasoning.
3. **Knowledge acquisition subsystem:** A subsystem to help experts build knowledge bases. Collecting knowledge needed to solve problems and build the knowledge base continues to be the biggest bottleneck in building expert systems.
4. **Explanation subsystem:** A subsystem that explains the system's actions. The explanation can range from how the final or intermediate solutions were arrived at to justifying the need for additional data.
5. **User interface:** The means of communication with the user. The user interface is generally not a part of the ES technology, and was not given much attention in the past. However, it is

now widely accepted that the user interface can make a critical difference in the perceived utility of a system regardless of the system's performance.

6.

15.What are the essential characteristics that is need for a robot?

Sensing First of all your robot would have to be able to sense its surroundings.

Movement A robot needs to be able to move around its environment

Energy A robot needs to be able to power itself

Intelligence A robot needs some kind of "smarts."

11 Marks

1. Explain Minimax Search Procedure - Adding Alpha-Beta Cutoffs In Game Playing?
2. Explain Expert System Its Expert System Shells And Knowledge Acquisition.
3. Expert System Architecture
4. Explain Robotics And Its Hardware, Perception
5. Explain Swarm Intelligent Systems .(or) What Is Ant Colony System, Its Application And Working.
6. Write Short Notes On:
 - a. Expert System Shell.
 - b. Knowledge System Building Tools.
7. Discuss Alpha- Beta Cutoffs.
8. Explain Rule Based System Architecture.
9. Describe The Non Production System
10. Write Short Notes On Knowledge Acquisition.